Imperial College London

Department of Electronic and Electrical Engineering

Examinations 2024

Computational Sensing and Imaging

MSc and EEE/EIE PART IV

May, 2024 • Exam Duration: Three Hours

All Questions are Compulsory

Where unavoidable, make reasonable assumptions for your solutions. Explain your assumptions in the answer script.

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Questions [100 Marks]

1) What is Computational Sensing?

How is it different from signal or image processing? **[2 Marks]**

Computational imaging is the process of indirectly forming images from measurements using algorithms that rely on a significant amount of computing. In contrast to traditional imaging, computational imaging involves designing the forward model and integrating the sensing system.

2) Provide 2 examples of computational sensing based systems. **[2 Marks]**

1. A modified uniformly redundant array (MURA) is a type of mask used in coded aperture imaging. 2. Time-resolved imaging. 3. One-bit sigma-delta sensing. 4. MRI imaging.

3) Consider a frequency estimation problem:

$$f(x) = c_1\alpha^x + c_2\beta^x, \qquad x = 0, 1, 2, 3.$$

Establish the following relation,

$$\begin{bmatrix} f(0) & f(1) \\ f(1) & f(2) \end{bmatrix} \begin{bmatrix} \alpha\beta \\ -\alpha-\beta \end{bmatrix} = -\begin{bmatrix} f(2) \\ f(3) \end{bmatrix}$$

Hint: You can see that this is related to Prony's method extensively covered in the class. 1) Use the annihilation polynomial in Prony's method: $h(z) = (z-\alpha)(z-\beta)$. Note that, $(f * h)[n] = 0$ where $\{h[n]\}_{n=0}^{2} = \{\lambda_0, \lambda_1, 1\}$, and 2) Utilize the fact that $h(\alpha) = h(\beta) = 0$. **[8 Marks]**

Let $h(z) = (z-\alpha)(z-\beta) = z^2 + \lambda_1 z + \lambda_0$ where $\lambda_1 = -\alpha - \beta$ and $\lambda_0 = \alpha\beta$. Since $h(\alpha) = h(\beta) = 0$, we have

$$\begin{cases} c_1 h(\alpha) + c_2 h(\beta) = 0 \\ c_1\alpha h(\alpha) + c_2\beta h(\beta) = 0 \end{cases}. \tag{1}$$

Since $h(z) = (z-\alpha)(z-\beta) = z^2 + \lambda_1 z + \lambda_0$, then, we have that,

$$\begin{cases} c_1(\alpha^2 + \lambda_1\alpha + \lambda_0) + c_2(\beta^2 + \lambda_1\beta + \lambda_0) = 0 \\ c_1(\alpha^3 + \lambda_1\alpha^2 + \lambda_0\alpha) + c_2(\beta^3 + \lambda_1\beta^2 + \lambda_0\beta) = 0 \end{cases}. \tag{2}$$

Reorganizing terms in λ_0 and λ_1 leads to

$$\begin{cases} \lambda_0(c_1 + c_2) + \lambda_1(c_1\alpha + c_2\beta) + c_1\alpha^2 + c_2\beta^2 = 0 \\ \lambda_0(c_1\alpha + c_2\beta) + \lambda_1(c_1\alpha^2 + c_2\beta^2) + c_1\alpha^3 + c_2\beta^3 = 0 \end{cases}. \tag{3}$$

Hence, the above system of linear equations can be written as,

$$\begin{bmatrix} f(0), & f(1) \\ f(1), & f(2) \end{bmatrix} \begin{bmatrix} \lambda_0 \\ \lambda_1 \end{bmatrix} = -\begin{bmatrix} f(2) \\ f(3) \end{bmatrix}$$

which completes the proof of the claim.

4) Consider the following constrained least squares problem of the form:

$$\min_{\mathbf{x}} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2, \quad \text{such that} \quad \mathbf{C}\mathbf{x} = \mathbf{b}. \tag{4}$$

Show the optimal solution is:

$$\mathbf{x}^* = (\mathbf{H}^\top\mathbf{H})^{-1}\left(\mathbf{H}^\top\mathbf{y} - \mathbf{C}^\top(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{C}^\top)^{-1}(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{y} - \mathbf{b})\right).$$

[8 Marks]

Define the Lagrangian,

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_2^2 + \boldsymbol{\mu}^\top(\mathbf{C}\mathbf{x} - \mathbf{b}).$$

The derivatives are:

$$\frac{\partial}{\partial \mathbf{x}}\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = 2\mathbf{H}^\top(\mathbf{H}\mathbf{x} - \mathbf{y}) + \mathbf{C}^\top\boldsymbol{\mu}$$
$$\frac{\partial}{\partial \boldsymbol{\mu}}\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = \mathbf{C}\mathbf{x} - \mathbf{b}$$

Setting the derivatives to zero,

$$\frac{\partial}{\partial \mathbf{x}}\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = 0 \Longrightarrow \mathbf{x} = (\mathbf{H}^\top\mathbf{H})^{-1}(\mathbf{H}^\top\mathbf{y} - \mathbf{0.5}\mathbf{C}^\top\boldsymbol{\mu}) \tag{5}$$
$$\frac{\partial}{\partial \boldsymbol{\mu}}\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = 0 \Longrightarrow \mathbf{C}\mathbf{x} = \mathbf{b} \tag{6}$$

Multiplying (5) on the left by $\mathbf{C}$ gives $\mathbf{C}\mathbf{x}$, which from (6) is $\mathbf{b}$, so we have that

$$\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}(\mathbf{H}^\top\mathbf{y} - \frac{\mathbf{1}}{\mathbf{2}}\mathbf{C}^\top\boldsymbol{\mu}) = \mathbf{b}$$

or, expanding,

$$\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{y} - \frac{\mathbf{1}}{\mathbf{2}}\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{C}^\top\boldsymbol{\mu}) = \mathbf{b}.$$

Solving for $\boldsymbol{\mu}$ gives

$$\boldsymbol{\mu} = 2\left(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{C}^\top\right)^{-1}\left(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{y} - \mathbf{b}\right)$$

Plugging $\boldsymbol{\mu}$ into (6) gives

$$\mathbf{x}^* = (\mathbf{H}^\top\mathbf{H})^{-1}\left(\mathbf{H}^\top\mathbf{y} - \mathbf{C}^\top(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{C}^\top)^{-1}(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{y} - \mathbf{b})\right)$$

Let us verify that $\mathbf{x}$ in this formula does in fact satisfy $\mathbf{C}\mathbf{x} = \mathbf{b}$,

$$\begin{aligned}
\mathbf{C}\mathbf{x}^* &= \mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\left(\mathbf{H}^\top\mathbf{y} - \mathbf{C}^\top(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{C}^\top)^{-1}(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{y} - \mathbf{b})\right) \\
&= \mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{y} - \mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{C}^\top\left(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{C}^\top\right)^{-1}(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{y} - \mathbf{b}) \\
&= \mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{y} - (\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{y} - \mathbf{b}) = \mathbf{b}
\end{aligned}$$

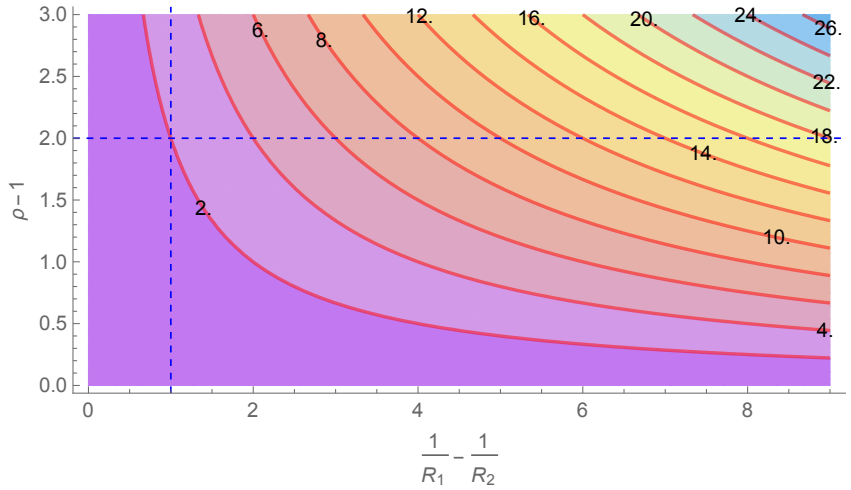
So,

$\min_{\mathbf{x}} \|\mathbf{y}-\mathbf{H}\mathbf{x}\|_2^2 \quad \text{s.t.} \quad \mathbf{C}\mathbf{x}=\mathbf{b} \quad \Longrightarrow$
$\mathbf{x}^* = (\mathbf{H}^\top\mathbf{H})^{-1}\left(\mathbf{H}^\top\mathbf{y} - \mathbf{C}^\top(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{C}^\top)^{-1}(\mathbf{C}(\mathbf{H}^\top\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{y}-\mathbf{b})\right).$

5) Let $\rho = n_2/n_1$, the ratio of refractive indices.
The well known Lens Maker's equation is given by,

$$(\rho-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)=\frac{1}{f}$$

and for a certain imaging system, the operation range is described in the figure below.



(a) Explain the solid red lines in this figure. **[2 Marks]**

(b) Also, given materials such that $\rho \geqslant 3$, discuss the constraints on $\{R_1, R_2\}$ so that an effective focal length of half can be attained by the system. **[3 Marks]**

(a) The solid red lines are equipotential lines on which the points share the same focal length. Moving towards right and top, the focal length decreases ($\frac{1}{f}$ increases).

(b) An effective focal length of half means that $\frac{1}{f}=2$ which corresponds to the red line at the leftmost. The constraints $\rho \geqslant 3$ requires that $\{R_1, R_2\}$ can only take the values that are above the horizontal blue line ($\rho = 3$). Hence, $\{R_1, R_2\}$ should take points on the leftmost red line above the horizontal blue line.

6) Sparse recovery beyond soft thresholding **[15 Marks]**
In the context of sparse recovery, the cost function

$$\mathcal{C}_\lambda(\mathbf{f}) = \|\mathbf{g}-\mathbf{f}\|_2^2 + \lambda\|\mathbf{f}\|_{\ell_1} = \sum_{n=1}^{N}(g_n-f_n)^2 + \lambda|f_n|.$$

can be generalized in that the ℓ_1–norm can be replaced by a pointwise nonlinearity defined by function ξ, leading to

$$\mathcal{C}_\lambda(\mathbf{f}) = \|\mathbf{g}-\mathbf{f}\|_2^2 + \lambda\xi(\mathbf{f})$$

provided that ξ is differentiable. The core idea is to use a ξ that mimics the ℓ_1–norm.

(a) Explain why we need ξ to be differentiable.

As shown in the next step, differentiability ensures that we can minimize the cost function defined in terms of ξ.

(b) Suppose that we define

$$\xi(\mathbf{f}) = \frac{1}{\alpha}\log(1+\alpha|\mathbf{f}|), \quad \alpha > 0.$$

For one-dimensional and two-dimensional $\mathbf{f}$ plot $\xi(\mathbf{f})$ and compare it with the ℓ_1–norm of $\mathbf{f}$.

In analogy to the minimizer given by, $\mathbf{f}^\star = \text{soft}_\lambda(\mathbf{g})$, what is the minimizer of the cost function with $\xi(\mathbf{f})$ defined as previously?

The plots for the cost function are shown in Figure **??**. We can write the cost-function to be minimized as:

$$\mathcal{C}_\lambda(\mathbf{f}) = \|\mathbf{g}-\mathbf{f}\|_2^2 + \lambda\frac{1}{\alpha}\log(1+\alpha|\mathbf{f}|) = \sum_{n=1}^{N}(g_n-f_n)^2 + \lambda\frac{1}{\alpha}\log(1+\alpha|f_n|).$$

To minimize the same, we set $\frac{\partial\mathcal{C}_\lambda(\mathbf{f})}{\partial\mathbf{f}}$ to be zero which gives,

$$\frac{\partial\mathcal{C}_\lambda(\mathbf{f})}{\partial\mathbf{f}} = \mathbf{0} \Rightarrow \mathbf{g} = \mathbf{f} + \lambda\frac{\partial\xi(\mathbf{f})}{\partial\mathbf{f}}.$$

To compute this, we first compute (element-wise),

$$\frac{\partial\xi(f_n)}{\partial f_n} = \begin{cases} \frac{1}{1+\alpha f_n} & f_n > 0 \\ -\frac{1}{1-\alpha f_n} & f_n < 0 \end{cases} = \frac{1}{1+\alpha|f_n|}\text{sgn}(f_n).$$

And hence, we finally obtain,

$$\frac{\partial\mathcal{C}_\lambda(\mathbf{f})}{\partial\mathbf{f}} = \mathbf{0} \Rightarrow g_n = f_n + \frac{\lambda}{1+\alpha|f_n|}\text{sgn}(f_n).$$

To solve for $\mathbf{f}$, we proceed as follows,

$$g_n \geqslant \lambda, \qquad g_n = f_n + \frac{\lambda}{1+\alpha f_n} \Leftrightarrow (g_n-f_n)(1+\alpha f_n) - \lambda = 0.$$

This is a quadratic equation in variable f_n with coefficients a, b, c given as,

$$\underbrace{\alpha}_{a} f_n^2 + \underbrace{(1-\alpha g_n)}_{b} f_n + \underbrace{(\lambda - g_n)}_{c} = 0$$

and its solution is given by,

$$f_n = \frac{1}{2\alpha}\left((\alpha g_n - 1) + \sqrt{(\alpha g_n - 1)^2 - 4\alpha(\lambda - g_n)}\right), \qquad g_n \geqslant \lambda.$$

From these calculations we see that the $\mathsf{soft}_\lambda(\mathbf{g})$ version for $\xi(\cdot)$ is,

$$\begin{cases} \frac{1}{2\alpha}\left((\alpha\mathbf{g}-\mathbf{1}) + \sqrt{(\alpha\mathbf{g}-\mathbf{1})^2 - 4\alpha(\lambda\mathbf{1}-\mathbf{g})}\right) & |g_n| \geqslant \lambda \\ 0 & |g_n| \leqslant \lambda \end{cases}$$

where $\mathbf{1}$ is an all-ones vector of size N.

7) (a) What is the qualitative and mathematical definition of Plenoptic Function? Identify the different variables of the function. **[2 Marks]**

(b) Based on (a), explain (mathematically) how a smart-phone camera produces a 2D photograph from a plenoptic function. **[3 Marks]**

(c) Map each parameter in the plenoptic function (a) to an imaging modality. **[5 Marks]**

(a) The plenoptic function describes the intensity of each light ray in the world as a function of visual angle, wavelength, time, and viewing position. Mathematically, it is described by $P = P(\mathbf{r}, \theta, \phi, \lambda, t)$.

(b) Assume monochromatic and time-invariant, the plenoptic function has 5 variables. The model is further reduced to 4D as the radiance along a ray remains constant throughout its length. Then, the 2D photography is produced by projecting 4D light field into a 2D plane as $I(s,t) = \iint P(u,v,s,t)dudv$.

(c) $\mathbf{r}$, imaging at position $\mathbf{r}$; θ, ϕ, imaging angle; λ, imaging using light with wavelength λ (Hyper-spectral imaging); t, imaging at the time instance t (time-resolved imaging).

8) Noise Shaping Based Reconstruction.

Given a bandlimited function $f(t)$, in one-bit sampling, the samples $q[n] \in \{-1, 1\}$ are given by,

$$\begin{cases} q[n] & = \operatorname{sgn}(u[n-1] + f[n]) \\ u[n] & = u[n-1] + f[n] - q[n] \end{cases}.$$

(a) Re-write the one-bit samples as a single equation in terms of f and u. **[3 Marks]**

(b) From above, derive an equivalent expression in the Fourier Domain. **[2 Marks]**

(c) Show how an approach similar to Shannon's interpolation formula (low-pass filtering) can be used for reconstruction from one-bit samples. **[5 Marks]**

(a) From $u[n] = u[n-1] + f[n] - q[n]$, we have that,

$$q[n] = f[n] - (u[n] - u[n-1]) = f[n] - \Delta u[n]. \tag{7}$$

(b) Notice that,$q[n]$ can be decomposed as f and Δu, where f is low-pass and Δu is high-pass.

(c) With sufficient oversampling, $\widehat{q} \approx \widehat{f}$ in the low-pass region. This is because the cut-off frequency of Δu is larger than the bandwidth of f, provided that the oversampling rate is sufficiently large. Then, performing low-pass filtering on $q[n]$ leads to a sufficiently accurate reconstruction of f, for which the approximation error is determined by the oversampling rate.

9) In a certain time-of-flight system, the emitted signal is give by $p(t) = e^{-\frac{t^2}{2\sigma^2}}$ (Gaussian Excitation). Suppose the scene response function consists of a single reflection with reflectivity Γ and the object distance $d = \frac{ct_0}{2}$ where c is the light speed.

Use $\int_{\mathbb{R}} e^{-\frac{t^2}{2\sigma^2}} dt = \sqrt{2\pi}\sigma$ to simplify your computations.

(a) Write down the expression of the reflected signal $r(t)$. **[1 Marks]**

(b) Write down the cross-correlation signal $f(\tau) = \int r\,(t)\, p(t+\tau)dt$.

[4 Marks]

(c) Explain why the (above) cross-correlation technique is sub-optimal for resolving close-by reflections.

[4 Marks]

(d) For a single reflection, explain how sparse deconvolution method can be used for depth recovery. **[3 Marks]**

(a) The reflected signal is given by $r(t) = \Gamma e^{-\frac{(t-t_0)^2}{2\sigma^2}}$.

(b) The cross-correlation signal is given by

$$
\begin{aligned}
f(t) &= \int_{\mathbb{R}} r\,(t)\, p(t+\tau)dt = \int_{\mathbb{R}} \Gamma e^{-\frac{(t-t_0)^2}{2\sigma^2}} e^{-\frac{(t+\tau)^2}{2\sigma^2}} dt \\
&= \Gamma \int_{\mathbb{R}} e^{-\frac{(t-t_0)^2+(t+\tau)^2}{2\sigma^2}} dt = \Gamma \int_{\mathbb{R}} e^{-\frac{2\left(t^2+(\tau-t_0)t+\left(\frac{\tau-t_0}{2}\right)^2\right)+t_0^2+\tau^2-\frac{(\tau-t_0)^2}{2}}{2\sigma^2}} dt \\
&= \Gamma \int_{\mathbb{R}} e^{-\frac{2\left(t-\left(\frac{\tau-t_0}{2}\right)\right)^2+\frac{(\tau+t_0)^2}{2}}{2\sigma^2}} dt = \Gamma e^{-\frac{(\tau+t_0)^2}{4\sigma^2}} \int_{\mathbb{R}} e^{-\frac{2\left(t-\left(\frac{\tau-t_0}{2}\right)\right)^2}{2\sigma^2}} dt = \sqrt{\pi}\sigma\Gamma e^{-\frac{(\tau+t_0)^2}{4\sigma^2}}.
\end{aligned}
$$

(c) From the expressions of $p(t)$ and $f(t)$, we can clearly see that the width of the Gaussian pulse expands from σ of p to $\sqrt{2}\sigma$ of f. This expanding effect decreases the temporal resolution of time-delay estimation when locating peaks of the cross-correlation signal $f(t)$.

(d) Notice that, the reflected signal $r(t)$ can be expressed using the emitted signal $p(t)$,

$$r(t) = \Gamma p(t - t_0). \tag{8}$$

Hence, in Fourier-domain, we have that,

$$R(\omega) = \Gamma P(\omega) e^{-\jmath\omega t_0} \quad \Longrightarrow H(\omega) = \frac{R(\omega)}{P(\omega)} = \Gamma e^{-\jmath\omega t_0} \tag{9}$$

for which the time-delay t_0 can be estimated from $H(\omega)$ utilizing the Prony's method or high-resolution frequency estimation techniques.

10) Real-world signals are

(a) analog in amplitude,

(b) quantized in amplitude,

(c) vary continuously in time,

(d) are discrete in time.

(Multiple answers are possible.) **[2 Marks]**

Real-world signals are analog in amplitude and vary continuously in time.

11) How is a discrete signal scaled so that the sampling frequency satisfies $f_s = 1$? If the original analog signal has bandwidth Ω rad/s the scaled samples represent samples from an analog signal of what bandwidth? **[2 Marks]**

To achieve the desired scaling all frequencies are divided by $f_s = 1/T$, where T is the sampling period. This has the effect of dividing time samples by T. In the analog domain, the effect is of dividing the bandwidth by $1/T$ yielding the bandwidth $T\Omega$. When sampling at Nyquist rate, this results in bandwidth π rad/s.

12) In the context of signal/image denoising, people have developed filtering and thresholding techniques. Suppose a noisy signal $y[n]$ is given by $y[n] = x[n] + w[n], n = 0, \cdots, N-1$ where the original signal $x[n]$ is a finite sum of complex exponentials $x[n] = \sum_{k=0}^{K-1} c_k e^{\jmath \omega_k n}$ and $\max_k |\omega_k| \leqslant \Omega$. $w[n]$ is additive white Gaussian noise. Noting that $x[n]$ is sparse in Fourier domain, then the denoising problem can be formulated as

$$\min_{\widehat{\mathbf{x}}} \|\widehat{\mathbf{y}} - \widehat{\mathbf{x}}\|_2^2 + \lambda \|\widehat{\mathbf{x}}\|_{\ell_1}. \tag{10}$$

where $\widehat{\mathbf{x}}$ and $\widehat{\mathbf{x}}$ are the DFT of $\mathbf{x}$ and $\mathbf{y}$, respectively. Utilizing the soft-thresholding operator, write down the sparse recovery algorithm to solve (10) and explain the performance difference compared to the low-pass filtering methods.

[5 Marks]

Given the signal model $y[n] = x[n] + w[n]$, the recovery problem can be formulated as

$$\min_{\widehat{\mathbf{x}}} \|\widehat{\mathbf{y}} - \widehat{\mathbf{x}}\|_2^2 + \lambda \|\widehat{\mathbf{x}}\|_{\ell_1} \Longleftrightarrow \min_{\widehat{x}} \sum_{n=0}^{N-1} \left((\widehat{y}_n - \widehat{x}_n)^2 + \lambda |\widehat{x}_n| \right). \tag{11}$$

where λ is the penalty parameter and $\widehat{x}$ denotes the Dicrete-Fourier Transform of x. This leads to the optimal solution using the soft-thresholding operator

$$x^* = \text{IDFT}(\text{soft}_\lambda(\widehat{\mathbf{y}})) = \text{IDFT}\left(\text{sgn}(\widehat{\mathbf{y}}) \left(|\widehat{\mathbf{y}}| - \frac{\lambda}{2} \right) \mathbb{1}_{|\widehat{y}| > \frac{\lambda}{2}}(\widehat{\mathbf{y}}) \right). \tag{12}$$

The soft-thresholding operator defined in (12) completely annihilates the noise in the measurements $\widehat{y}[n]$ whenever $|\widehat{y}| \leqslant \frac{\lambda}{2}$. However, filtering operates in the Fourier domain and cannot eliminate the noise in the bandwidth $[-\Omega, \Omega]$. In other words, the background noise in bandwidth $[-\Omega, \Omega]$ remains after the filtering operation.

13) Based on the diagram shown in Fig. 1 ($m = 0, \cdots, M-1$), write down the forward model of the single-pixel imaging. Show that the single-pixel imaging problem can be solved using compressive sensing methods. **[5 Marks]**

The measurement at time instance m is given by $y[m] = \langle \mathbf{x}, \boldsymbol{\phi}_m \rangle$ where $\mathbf{x} \in \mathbb{R}^N$ is a sampled version of an object into N pixels and $\boldsymbol{\phi}_m$ is a test signal. Using a set of M test signals, the measurements can be expressed as

$$\begin{bmatrix} y[0] \\ y[1] \\ \vdots \\ y[M-1] \end{bmatrix} = \begin{bmatrix} \boldsymbol{\phi}_0 \\ \boldsymbol{\phi}_1 \\ \vdots \\ \boldsymbol{\phi}_{M-1} \end{bmatrix} \mathbf{x} = \boldsymbol{\Psi}\mathbf{x}. \tag{13}$$

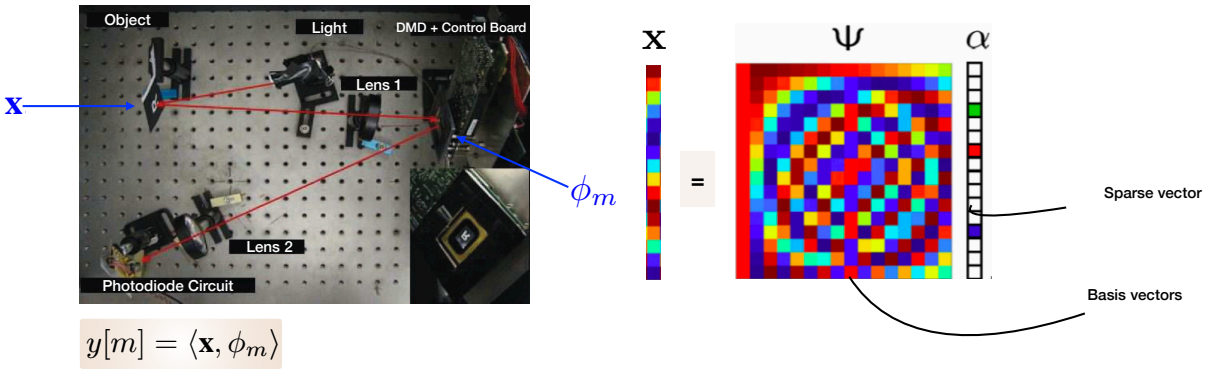


Fig. 1: Illustration of the single pixel imaging.

With the sparsity of the object image, we can write image $\mathbf{x}$ as $\mathbf{x} = \boldsymbol{\Psi}\boldsymbol{\alpha}$, where $\boldsymbol{\alpha}$ are the sparse coefficients that represent $\mathbf{x}$ using the atoms in $\boldsymbol{\Psi}$. Eventually, the measurements can be expressed as, $\mathbf{y} = \boldsymbol{\Psi}\boldsymbol{\Psi}\boldsymbol{\alpha}$ where $\boldsymbol{\alpha}$ are the sparse coefficients, which can be estimated via the following minimization problem

$$\mathbf{x}^* = \underset{\boldsymbol{\alpha}}{\operatorname{argmin}} \|\boldsymbol{\alpha}\|_{\ell_1} + \lambda \|\mathbf{y} - \boldsymbol{\Psi}\boldsymbol{\Psi}\boldsymbol{\alpha}\|_2^2 . \tag{14}$$

14) What is the main difference between multi-spectral and hyper-spectral imaging?

[3 Marks]

Multi-spectral imaging is not necessarily uniform bandwidth while the hyper-spectral imaging has uniform bandwidth.

15) Given two examples of imaging techniques with non-visible frequencies. **[3 Marks]**

(a) Wi-Fi imaging;
(b) Thermal imaging;
(c) X-ray imaging.

16) Compared to the conventional 2D photography, what is the significant feature of time-resolved imaging?

[2 Marks]

Furthermore, why it is difficult to achieve both temporal and spatial resolution in time-resolved imaging technologies?

[2 Marks]

Time-resolved imaging preserves the time variable of the plenoptic function which captures a scene-dependent time profile at a specific time resolution, usually at the magnitude of nanoseconds. The temporal resolution is the number of pixels at the ToF sensor while the spatial resolution describes the time scale of exposure. Due to the time profile at each pixel, the data size explodes largely as the number of pixels increases or the time scale increases.

17) Consider a linear regression problem $\min_c \|\mathbf{y} - c\mathbf{x}\|_2^2$ where $\mathbf{x}, \mathbf{y} \in \mathbb{Z}^N$.
Show that the minimum least-squares error is given by

$$\|\mathbf{y}\|_2^2 - \frac{|\mathbf{x}^{\mathsf{H}}\mathbf{y}|^2}{\|\mathbf{x}\|_2^2}.$$

[4 Marks]

First, we know that the optimal linear coefficient c is given by

$$c^* = (\mathbf{x}^{\mathsf{H}}\mathbf{x})^{-1}\mathbf{x}^{\mathsf{H}}\mathbf{y}. \tag{15}$$

Then, the objective function at the minimizer c^* can be expressed as

$$J(c^*) = \|\mathbf{y} - c^*\mathbf{x}\|_2^2 = (\mathbf{y} - c^*\mathbf{x})^{\mathsf{H}}(\mathbf{y} - c^*\mathbf{x}) \tag{16}$$

$$= \frac{|\mathbf{x}^{\mathsf{H}}\mathbf{y}|^2}{\|\mathbf{x}\|^2} - \frac{|\mathbf{x}^{\mathsf{H}}\mathbf{y}|^2}{\|\mathbf{x}\|^2} - \frac{|\mathbf{x}^{\mathsf{H}}\mathbf{y}|^2}{\|\mathbf{x}\|^2} + \|\mathbf{y}\|^2 = \|\mathbf{y}\|_2^2 - \frac{|\mathbf{x}^{\mathsf{H}}\mathbf{y}|^2}{\|\mathbf{x}\|_2^2} \tag{17}$$